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Multilevel Stewardship Intervention for Use of Anticoagulation-Antiplatelet Therapy

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IMPORTANCE Antiplatelet medications are overprescribed in patients taking direct oral anticoagulants (DOACs), increasing their risk of major bleeding. The utility of potentially scalable antithrombotic stewardship approaches remains unknown.

OBJECTIVE To evaluate a multicomponent antithrombotic stewardship initiative to reduce unnecessary antiplatelet use in patients prescribed DOACs.

DESIGN, SETTING, AND PARTICIPANTS This quality improvement study used retrospective multiperiod comparative interrupted-time-series analysis from July 2020 to July 2023 to compare intervention and control sites. Participants were adults prescribed DOACs in the ambulatory setting. The interventions occurred in 7 Veterans Health Administration (VHA) health systems, while 128 other VHA health systems served as controls. Data were analyzed from July 2023 to March 2026.

INTERVENTION In stage 1, lasting 9 months, intervention sites implemented educational outreach to clinicians and patients and changes to the electronic health record system. In stage 2, lasting 16 months, a clinical pharmacist–facing electronic flag identifying patients receiving antiplatelet therapy was added to a widely used electronic dashboard.

MAIN OUTCOMES AND MEASURES Monthly site-level percentage of patients prescribed antiplatelet medications. The summary measure was the difference in the semiannual change in the outcome for intervention compared with control sites, controlling for preintervention trends. Subgroup analyses were performed based on antiplatelet indication.

RESULTS This study found that preintervention antiplatelet use in patients prescribed DOACs was 26.1% (95% CI, 26.0%-26.1%) in the 7 intervention sites (27 588 patients; 704 females [2.6%]) and 30.1% (95% CI, 30.0%-30.2%) in 128 control sites (253 085 patients; 6481 females [2.6%]). Antiplatelet use decreased faster by an absolute −0.58 (95% CI, −0.95 to −0.22) percentage points (pp) per 6 months for intervention compared with control sites after the 2 interventions had been implemented. The initial set of interventions was associated with an absolute −0.29 (95% CI, −0.61 to 0.04) pp change per 6 months and later augmentation with the electronic flag was associated with an absolute −0.29 (95% CI, −0.61 to 0.03) pp change per 6 months. The combined interventions were associated with the greatest reduction in the subgroup of patients with stable coronary artery disease (absolute −2.1 [95% CI, −3.0 to −1.2] pp per 6 months, equivalent to a −5.5% additional change compared with the baseline prevalence in this group), for whom antiplatelet deimplementation is likely appropriate.

CONCLUSIONS AND RELEVANCE This study found that the combined interventions were associated with a clinically meaningful reduction in potentially harmful combination antithrombotic therapy. The initial educational outreach and changes to the electronic health record and later augmentation with the electronic flag had additive effects, highlighting the importance of multilevel interventions to speed adoption of evidence-based antithrombotic prescribing.

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Invited Commentary page 1009

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Anticoagulants are the leading cause of adverse drug events resulting in emergency department visits, most often causing bleeding.[1] Nearly one-third of patients treated with oral anticoagulants (OACs) also take antiplatelet medications (eg, low-dose [not more than 100 mg] aspirin or a P2Y12 inhibitor).[2] A meta-analysis of 5 randomized clinical trials[3] found that, for patients with coronary artery disease (CAD) requiring long-term anticoagulation, combining antiplatelet therapy with OACs increases the risk of major bleeding without reducing thromboembolism compared with OAC monotherapy. Accordingly, indications for combined therapeutic dose anticoagulant-antiplatelet therapy have narrowed. The population who most benefit are patients with recent acute coronary syndrome, revascularization, or percutaneous coronary intervention plus an indication for anticoagulation.[4,5] It is thus estimated that up to 90% of patients receiving combination therapy may not have a current indication for the antiplatelet medication.[6,7]

To address such gaps, antithrombotic stewardship has become an increasingly accepted approach for systematically ensuring evidence-based use of anticoagulant and antiplatelet medications.[8] However, multiple barriers impede antiplatelet deimplementation: fear of precipitating thromboembolic events, uncertainty about appropriate indications, ambiguity about the responsible clinician, and unfamiliarity with relevant clinical trials.[9,10] Yet, the best strategies for overcoming these barriers are unknown. One potential strategy is academic detailing (AD) (ie, educational outreach and coaching of clinicians), but it has not been evaluated for antithrombotic deimplementation.[11] Another pharmacist-facing strategy is electronic population management tools (ie, dashboards).[12,13] Combining these 2 approaches has theoretical appeal: both clinicians and clinical pharmacists can influence prescribing practices. Furthermore, a clinician who has received AD may be more responsive if later contacted by a clinical pharmacist responding to an electronic flag.

Here, we retrospectively evaluated a multicomponent antithrombotic stewardship initiative to reduce unnecessary antiplatelet use in patients prescribed direct OACs (DOACs). The 2-stage initiative, which has been previously described,[14] first used educational outreach and changes to the electronic health record (EHR) system and later an electronic flag in a dashboard to alert clinical pharmacists to combination DOAC-antiplatelet use. This design enabled us to estimate the overall impact and the effect of each stage. Given the known harms of inappropriate combination therapy, the study aims to identify potentially scalable approaches to antithrombotic stewardship.

Key Points

Question Is use of a multicomponent antithrombotic stewardship initiative associated with reduced antiplatelet use in patients prescribed direct oral anticoagulants?

Findings In this quality improvement study including 7 intervention sites (27 588 patients) and 128 control sites (253 085 patients), the interventions were associated with a decrease in antiplatelet use by −0.6 percentage points per 6 months overall and −2.1 percentage points in patients with stable coronary artery disease.

Meaning Results of this study suggest that these scalable multilevel interventions could result in clinically meaningful reductions in potentially harmful combination antithrombotic therapy.

Methods

This quality improvement study used retrospective multiperiod comparative interrupted-time-series analysis from July 2020 to July 2023 to compare intervention and control sites. Data were analyzed from July 2023 to March 2026. The Transparent Reporting of Nonrandomized Designs (TREND) reporting guideline was followed in reporting this study.[15] This evaluation qualified as a nonresearch quality improvement activity conducted under the authority of the Veterans Health Administration (VHA) operations per the VHA Program Guide 1200.21 and thus did not require institutional review board review.

Setting and Participants

The VHA operates the largest integrated health care system in the US, encompassing more than 140 medical centers and 1100 outpatient clinics in 18 regions. Most medications are dispensed by integrated pharmacies, and OAC use is overseen by pharmacy teams primarily dedicated to anticoagulation management. These teams consistently use an internally developed electronic population health management tool, which flags potentially unsafe prescribing (eg, inappropriate kidney dosing).[16,17] Throughout the VHA clinical pharmacists have prescriptive authority within their clinical scope of practice, which did not vary between intervention and control sites.

The interventions occurred in Veterans Integrated Service Network (VISN) 8, comprising 7 health care systems and 90 outpatient clinics in Florida, south Georgia, Puerto Rico, and the US Virgin Islands. All other 128 VHA health systems served as controls, with the exception of 10 that had implemented similar initiatives or transitioned to a separate EHR system.

Each month, patients were eligible for analysis if receiving a VHA-dispensed DOAC on the first of the month. Patients were considered exposed to the intervention if the prescription originated from a VISN 8 medical center.

Data Sources

Patient demographic, comorbidity, medication, hospitalization, and procedure data were extracted from the VHA Corporate Data Warehouse, a centralized systemwide data repository. This included procedures and hospitalizations paid for by VHA but provided elsewhere.

Intervention

The pharmacy anticoagulation workgroup in VISN 8 developed the OAC-antiplatelet stewardship initiative. Details of the interventions are described in eMethods 1 and 2 in Supplement 1. Health system participation was voluntary. In stage 1, two principal intervention approaches, named according to standard nomenclature in implementation science,[18] were available. The first intervention approach was educational outreach visits. A clinical pharmacist trained in AD (D.P.) provided individual and group AD to clinicians, a widely used and effective change strategy in VHA.[11] In addition, the academic detailer met with local clinical pharmacists to empower them to further influence and collaborate with local clinicians.

The second intervention approach was changes to the EHR system. New language was embedded in electronic DOAC prior authorization requests and in templated clinical pharmacist consultation notes to prompt reconsideration of the need for antiplatelet medications. The workgroup also distributed templated letters that clinical pharmacists could send to non-VHA clinicians and their patients requesting they reconsider the need for antiplatelet therapy.

In stage 2, to remind clinical pharmacists, an electronic flag was activated in the DOAC population management tool that identified patients receiving antiplatelet medications. Based on the electronic medication list, the flag alerted pharmacists to any antiplatelet medication dispensed by VHA or previously entered as a non–VHA-dispensed medication. The population management tool was primarily used by anticoagulation-focused clinical pharmacists during clinical time reserved for population health activities. The flag prompted a medication review, outreach to the patient or clinicians if necessary, and potentially deprescribing by the clinical pharmacist if within their scope of practice.

Rollout of stage 1 interventions occurred from January to June 2021 and was completed by July 2021, and the stage 2 electronic flag was implemented in April 2022. This created 4 periods of the study: (1) preintervention (July-December 2020), (2) preparation and rollout of the initiative (January-June 2021), (3) stage 1 following implementation of educational outreach and changes to the electronic record system (July 2021-March 2022), and (4) stage 2 following implementation of the clinical pharmacist–facing electronic flag (April 2022-July 2023), during which the interventions from stage 1 continued.

Outcomes

The primary outcome was the site-month percentage of patients prescribed antiplatelet medications (aspirin or P2Y12 inhibitor; drug names are given in eTable 1 in Supplement 1). For VHA medications, patients were considered to be taking the antiplatelet medication from the dispensed date through a period of days equal to twice the dispensed days' supply.[19] For documented non-VHA medications (without refill dates), we assumed continued use until removal from the medication list. We also examined whether the intervention was associated with off-target changes in the use of proton pump inhibitors (PPIs) in patients receiving combination therapy, which reduces the risk of gastrointestinal bleeding, and nonsteroidal anti-inflammatory drugs (NSAIDs), which may cause gastrointestinal bleeding.

Clinical Subgroups

To evaluate for heterogeneity of treatment effects, we analyzed antiplatelet use in 4 hierarchical and mutually exclusive clinical subgroups based on the likely appropriateness of combination therapy. The first clinical subgroup was patients with acute coronary syndrome, coronary artery bypass graft, or percutaneous coronary intervention in the prior 12 months. Contemporaneous professional guidance suggested these patients receive up to 12 months of combination therapy.[4]

The second clinical subgroup was patients with peripheral artery disease or antiphospholipid syndrome. For this group, observational data are evolving and clinical trial evidence is lacking, so there are no strong guideline recommendations.[20,21]

The third clinical subgroup was patients with stable CAD, defined as having CAD but not belonging to group 1. For this group, concomitant antiplatelet medications are not routinely recommended.[4]

The fourth clinical subgroup consisted of patients with none of the above indications, for whom antiplatelet medications are inappropriate. Diagnosis and procedure codes used to define these groups are shown in eTable 2 in Supplement 1. We also examined for differential site effect estimates.

Statistical Analysis

Patient characteristics were examined for the last month of the preintervention period (December 2020). A 1-year lookback period was used to determine comorbidities and calculate the Charlson Comorbidity Index.[22]

For medication outcomes, we collapsed person-level monthly use data to the site-month level. We examined parallel trends in the site-month antiplatelet medication use data as a percentage between intervention and control sites in the preintervention period. We then fit a piecewise linear regression model to the preintervention period and stage 1 and 2 site-month antiplatelet medication use data as a percentage and calculated the effect estimate of the educational outreach with changes to the electronic record system, the clinical pharmacist–facing electronic flag, and the combination. eMethods 3 in Supplement 1 provides additional details. We excluded the rollout period for stage 1 from the analyses in keeping with similar published studies.[23] Robust SEs were used for estimation from serial monthly measurements; each month's denominator was included in the model as analytic weights to account for differing numbers of eligible patients by site month. The absolute effect estimate of each intervention was estimated as the linear incremental change in use per 6 months based on slope differences corresponding to each stage, while controlling for preintervention trends.

Since only patients taking antiplatelet medications were eligible for antiplatelet deimplementation, for each intervention, we also calculated the change in antiplatelet use as a percentage of all patients receiving combination therapy at the end of the preintervention period (calculated as 100 × absolute effect estimate for the intervention divided by the preintervention antiplatelet use prevalence). For each intervention stage, the slope and absolute level of use in the last month were estimated based on the model. For exploratory analyses, we examined intervention outcomes associated with the use of VHA-dispensed aspirin, P2Y12 inhibitors, and non-VHA aspirin by using medication subgroups as dependent variables in separate models.

Table 1. Baseline Patient Characteristics for Control and Intervention Sites[a]

Characteristic	Participants, No. (%)	
	Control (n = 253 085)	Intervention (n = 27 588)
Age, mean (SD), y	74.3 (10.1)	74.9 (10.2)
Sex		
Female	6481 (2.6)	704 (2.6)
Male	246 599 (97.4)	26 884 (97.4)
Race[b]		
Asian	1043 (0.4)	30 (0.1)
Black	26 391 (10.4)	2369 (8.6)
Native American/Alaskan Native	1335 (0.5)	74 (0.3)
Native Hawaiian/Pacific Islander	1644 (0.6)	256 (0.9)
White	205 878 (81.3)	23 183 (84.0)
Multiracial	1628 (0.6)	162 (0.6)
Missing	15 166 (6.0)	1514 (5.5)
Ethnicity[b]		
Hispanic	6041 (2.4)	2785 (10.1)
Not Hispanic	236 094 (93.3)	23 705 (85.9)
Missing		
Rurality		
Rural	20 816 (8.2)	0 (0.0)
Urban	228 455 (90.3)	27 588 (100.0)
Missing	3814 (1.5)	0 (0.0)
Smoking status		
Current smoker	30 465 (12.0)	2783 (10.1)
Former smoker	122 194 (48.3)	13 573 (49.2)
Never smoker	79 966 (31.6)	9690 (35.1)
Unknown	20 460 (8.1)	1542 (5.6)
Medical history	10 950 (4.3)	1098 (4.0)
Atrial fibrillation	176 521 (69.7)	20 789 (75.4)
VTE	53 523 (21.1)	5840 (21.2)
Valve replacement	6159 (2.4)	745 (2.7)
Coronary artery disease	71 077 (28.1)	8681 (31.5)
Peripheral artery disease	45 416 (17.9)	5156 (18.7)
Stroke	27 494 (10.9)	3423 (12.4)
MACE in prior year	4145 (1.6)	473 (1.7)
APL	2443 (1.0)	243 (0.9)
COPD	56 788 (22.4)	6355 (23.0)
Hypertension	168 558 (66.6)	19 441 (70.5)
Cirrhosis	155 (0.1)	20 (0.1)
Peptic ulcer disease	1685 (0.7)	246 (0.9)
Diabetes	95 579 (37.8)	10 433 (37.8)
Charlson Comorbidity Index		
0	103 475 (40.9)	10 230 (37.1)
1	39 650 (15.7)	4202 (15.2)
≥2	109 960 (43.4)	13 156 (47.7)
DOAC prescribed		
Apixaban	169 160 (66.8)	13 601 (49.3)
Dabigatran	18 481 (7.3)	7131 (25.8)
Edoxaban	866 (0.3)	154 (0.6)
Rivaroxaban	64 578 (25.5)	6702 (24.3)
Antiplatelet use		
P2Y12 inhibitor only	12 573 (5.0)	1256 (4.6)
Aspirin only	60 091 (23.7)	5693 (20.6)
P2Y12 + aspirin	3576 (1.4)	245 (0.9)
None	176 845 (69.9)	20 394 (73.9)
Other medications		
PPI	76 784 (30.3)	8082 (29.3)
NSAID	7261 (2.9)	434 (1.6)

Abbreviations: APL, antiphospholipid syndrome; COPD, chronic obstructive pulmonary disease; DOAC, direct oral anticoagulant; MACE, major adverse coronary event, inclusive of percutaneous coronary intervention, acute coronary syndrome, or coronary artery bypass graft; NSAID, nonsteroidal anti-inflammatory drug; PPI, proton pump inhibitor; VTE, venous thromboembolism.

[a] The lookback period for Charlson Comorbidity Index score was 12 months.[22] Rurality was defined using the US Department of Agriculture definition, with the use of the Rural-Urban Commuting Area Codes framework based on the census tract of the patient's home address. Patient characteristics were determined for patients receiving DOACs in the month prior to the start of the rollout of stage 1 interventions (December 2020).

[b] Race and ethnicity data were derived from administrative records based on patient self-identification.

Analyses were performed using Stata version 18 (StataCorp LLC). Statistical significance was set at 2-sided $P < .05$.

Results

In the last month of the preintervention period, 27 588 patients (mean [SD] age, 74.9 [10.2] years; 97.4% male; 2.6% female) were prescribed DOACs in 7 intervention health systems and 253 085 patients (mean [SD] age, 74.3 [10.1] years; 97.4% male; 2.6% female) in 128 control health systems (**Table 1**). Patients at intervention sites had a higher prevalence of urban location (27 588 [100%] vs 228 455 [90.3%]), atrial fibrillation (20 789 [75.4%] vs 176 521 [69.7%]), dabigatran use (7131 [25.8%] vs 18 481 [7.3%]), and a lower prevalence of apixaban use (13 601 [49.3%] vs 169 160 [66.8%]).

Association of Interventions With Antiplatelet Use Overall

In the preintervention period, 30.1% (95% CI, 30.0%-30.2%) of patients in control sites and 26.1% (95% CI, 26.0%-26.1%) of patients in intervention sites used antiplatelet therapy; use in both groups was decreasing but at a slightly faster rate in intervention sites (**Table 2** and **Figure 1**). During stage 1, after implementation of the initial educational outreach and changes to the EHR system, the prevalence of antiplatelet use in intervention sites changed by −0.29 absolute percentage points (pp) (95% CI, −0.63 to 0.04 pp) per 6 months compared with control sites (Table 2). Similarly, the decrease in antiplatelet use further accelerated in intervention sites compared with control sites during stage 2 after additionally implementing the

Table 2. Slope of Change in Antiplatelet, NSAID, and PPI Use and Final-Month Prevalence of Use by Stage of Intervention for Intervention (VISN 8) and Control Sites, and Semi-Annual Changes Associated With Interventions[a]

	Preintervention period		Stage 1 (educational outreach and EHR changes)[b]		Stage 2 (clinical pharmacist–facing electronic flag)		Estimated effect (95% CI): Stage 1 interventions per 6 mo		Estimated effect (95% CI): Augmentation with stage 2 intervention per 6 mo		Estimated effect (95% CI): Combined interventions per 6 mo	
Medication class and site	Slope, PPC/6 mo (95% CI)	Mean level at end of period, % (95% CI)	Slope, PPC/6 mo (95% CI)	Mean level at end of period, % (95% CI)	Slope, PPC/6 mo (95% CI)	Mean level at end of period, % (95% CI)	APPC[c]	Change compared with VISN 8 baseline prevalence,[d] %	APPC[c]	Change compared with VISN 8 baseline prevalence,[d] %	APPC[c]	Change compared with VISN 8 baseline prevalence,[d] %
Antiplatelets												
Control[e]	−0.46 (−0.59 to −0.32)	30.1 (30.0 to 30.2)	−0.97 (−1.07 to −0.88)	28.4 (28.3 to 28.5)	−0.87 (−0.96 to −0.79)	26.1 (26.0 to 26.3)	−0.29 (−0.63 to 0.04)	−1.11 (−2.39 to 0.17)	−0.29 (−0.61 to 0.03)	−1.09 (−2.31 to 0.13)	−0.58 (−0.95 to −0.22)	−2.20 (−3.59 to −0.82)
VISN 8[e]	−0.94 (−1.18 to −0.71)	26.1 (26.0 to 26.1)	−1.75 (−1.94 to −2.18)	23.3 (23.2 to 23.4)	−1.94 (−2.18 to −1.71)	17.9 (17.6 to 18.2)						
NSAIDs												
Control	−0.04 (−0.10 to 0.03)	4.5 (4.5 to 4.5)	−0.16 (−0.19 to −0.14)	4.2 (4.2 to 4.2)	−0.09 (−0.11 to −0.07)	4.0 (4.0 to 4.0)	0.27 (0.09 to 0.46)	10.14 (3.24 to 17.04)	−0.13 (−0.24 to −0.02)	−4.81 (−8.88 to −0.74)	0.14 (−0.05 to 0.34)	5.33 (−1.84 to 12.50)
VISN 8	−0.11 (−0.27 to 0.05)	2.6 (2.6 to 2.7)	0.04 (−0.02 to 0.11)	2.6 (2.6 to 2.7)	−0.01 (−0.09 to 0.07)	2.5 (2.4 to 2.6)						
PPIs in patients at high risk[f]												
Control	0.10 (0.03 to 0.17)	32.8 (32.8 to 32.9)	−0.06 (−0.16 to 0.05)	32.9 (32.8 to 33.0)	−0.04 (−0.11 to 0.04)	32.8 (32.7 to 32.9)	−0.22 (−0.77 to 0.32)	−0.69 (−2.36 to 0.97)	0.30 (0.09 to 0.51)	0.92 (0.27 to 1.57)	0.08 (−0.46 to 0.61)	0.23 (−1.40 to 1.86)
VISN 8	0.01 (−0.50 to 0.52)	32.5 (32.3 to 32.6)	−0.37 (−0.50 to −0.24	32.3 (32.2 to 32.4)	−0.05 (−0.15 to 0.05)	32.2 (32.0 to 32.4)						

Abbreviations: APPC, absolute percentage-point change; DOAC, direct oral anticoagulant; EHR, electronic health record; NSAID, nonsteroidal anti-inflammatory drug; PPC, percentage-point change; PPI, proton pump inhibitor; VISN, Veterans Integrated Service Network.

[a] A piecewise regression model was fit with a 3-way interaction term for the stage 1 indicator by intervention site (VISN 8) indicator by continuous month and a 3-way interaction term for the stage 2 indicator by intervention site (VISN 8) indicator by continuous month, along with all lower interaction terms. These 3-way interaction terms were used to estimate the initial effect of the stage 1 interventions and the effect of the combined interventions. The augmenting effect of the clinical pharmacist–facing electronic flag alone was estimated as the difference between the rate of change associated with the combined interventions minus the rate of change associated with the stage 1 intervention alone, specifically as the difference between the two 3-way interaction terms. Period prevalence estimates were generated from regression models for the last month of each period. The slope in each stage was calculated using linear combinations of coefficients from the model.

[b] Once implemented, the stage 1 interventions continued throughout the remainder of the study period, including the period of the clinical pharmacist–facing electronic flag.

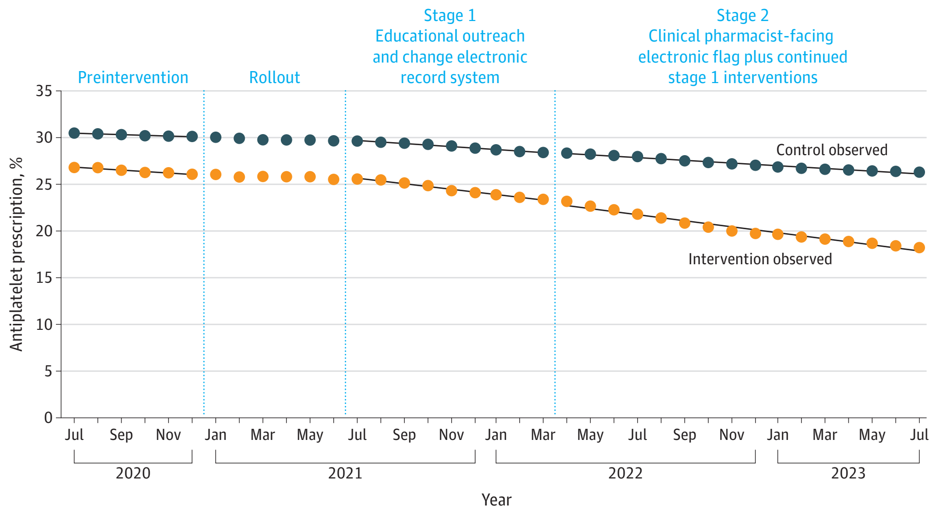
[c] Difference in slope (expressed as change per 6 months) for intervention compared with control sites during the specific intervention period, controlling for preintervention trends.

[d] Calculated as 100 × (absolute change/level in last month of the preintervention period in VISN 8). For example, the change in antiplatelet use compared with baseline associated with the combined interventions was calculated as 100 × −0.58/26.1 = −2.2%, meaning out of 100 patients with DOAC-antiplatelet use in the preintervention period in VISN 8, antiplatelet discontinuation for 2-3 patients per 6 months is attributable to the combined interventions by the end of stage 2.

[e] The mean number of patients prescribed DOACs per month in the preintervention period was 245 998 in control sites and 26 862 analyzed in VISN 8; in stage 1, 285 060 and 31 126, respectively; and in stage 2, 315 787 and 34 347, respectively.

[f] Patients were considered high risk for upper gastrointestinal bleeding if prescribed an antiplatelet medication or NSAID, in addition to a DOAC.

Figure 1. Line Graph Showing the Percentage of Patients Taking Antiplatelet Medications Per Month for Intervention and Control Sites



The intervention site was Veterans Integrated Service Network (VISN) 8, and control sites consisted of all other Veterans Health Administration sites (n = 128), with the exclusion of 10 sites that had previously used a similar antiplatelet safety intervention or had transitioned to a new electronic health record. The denominator includes all patients treated with direct oral anticoagulants. Fitted lines were created using predicted values from a piecewise regression model for each arm per month. The model was fit with a 3-way interaction term for the stage 1 indicator by intervention site (VISN 8) indicator by continuous month and a 3-way interaction term for the stage 2 indicator by intervention site (VISN 8) indicator by continuous month, along with all lower interaction terms. Data from the rollout period were excluded. **Table 2** shows the absolute changes in antiplatelet use associated with each intervention period. Once implemented, the stage 1 interventions continued throughout the remainder of the study period.

clinical pharmacist-facing electronic flag, equal to an additional −0.29 absolute pp (95% CI, −0.61 to 0.03 pp) change in antiplatelet use per 6 months beyond the initial effect of the stage 1 interventions. Overall, the outcome of the combined 2-stage intervention as seen during stage 2 was an acceleration in the decrease in antiplatelet use for intervention compared with control sites of −0.58 absolute pp (95% CI −0.95 to −0.22 pp) per 6 months, equivalent to an additional −2.2% change in DOAC-antiplatelet use compared with the preintervention antiplatelet prevalence. Similar reductions were seen for patients with VHA-dispensed aspirin and non-VHA aspirin (eTable 3 in Supplement 1). The combined interventions were not associated with changes in PPI use in patients at high risk or NSAIDs.

Heterogeneity of Effect Estimates by Clinical Subgroup

The combined interventions were associated with a −2.1 absolute pp (95% CI, −3.0 to −1.2 pp) change per 6 months in antiplatelet use for patients with stable CAD, equivalent to a 5.5% reduction compared with the preintervention antiplatelet prevalence for this group (**Figure 2**; eTable 4 in Supplement 1). The changes were not statistically significant for patients without a clinical indication for combination therapy, with peripheral artery disease or antiphospholipid syndrome, or for patients with CAD and coronary interventions or events in the prior year.

Heterogeneity of Effect Estimates by Medical Center

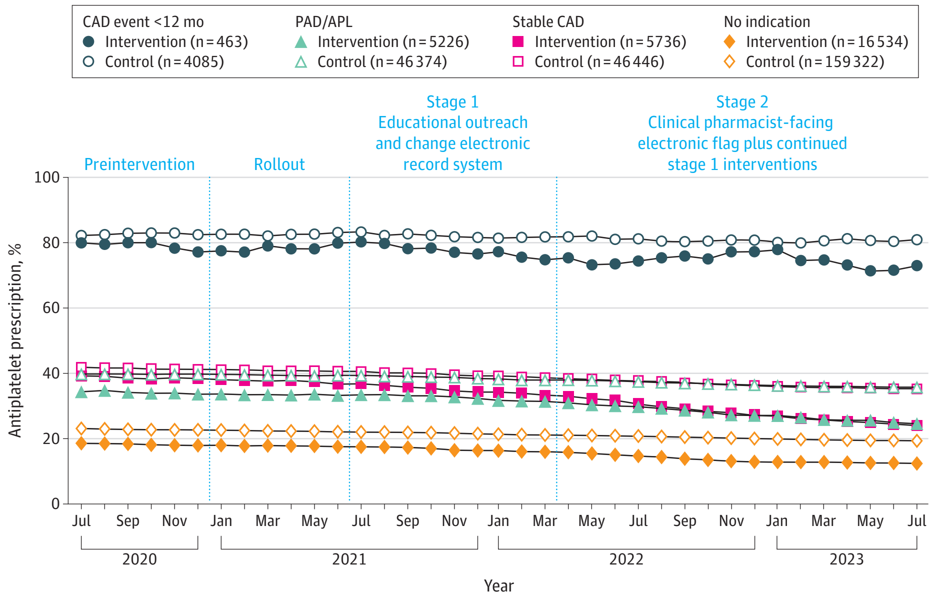
Compared with control sites, 3 of 7 intervention sites had significant reductions in antiplatelet use associated with the combined interventions, ranging from a −0.6 (95% CI, −1.0 to −0.3) absolute pp change per 6 months to −2.5 (95% CI, −3.4 to −1.4) absolute pp change per 6 months for intervention compared with control sites (**Figure 3**; eTable 5 in Supplement 1).

Discussion

In this multiperiod comparative interrupted-time-series analysis of VHA patients prescribed DOACs, we found that a large-scale multicomponent antithrombotic stewardship intervention in a regional network of health systems was associated with reduced use of antiplatelet therapy compared with control sites. This reduction was driven both by initial educational outreach (AD) and changes to the EHR system, and a subsequent clinical pharmacist-facing electronic flag to identify patients using antiplatelet therapy within a DOAC population management tool. The combined interventions were associated with the greatest reduction for patients with stable CAD, a group appropriate for antiplatelet deimplementation, in whom approximately 1 in 18 initial users had deimplementation per 6 months; changes in patients with peripheral artery disease or antiphospholipid syndrome (in whom definitive clinical guidance is lacking) and in patients with no indication for antiplatelet therapy were not statistically significant. Altogether, these findings suggest that the initial and later augmenting interventions had complementary roles in reducing antiplatelet therapy and represent scalable strategies to improve evidence-based antithrombotic prescribing, while highlighting opportunities to increase future impact.

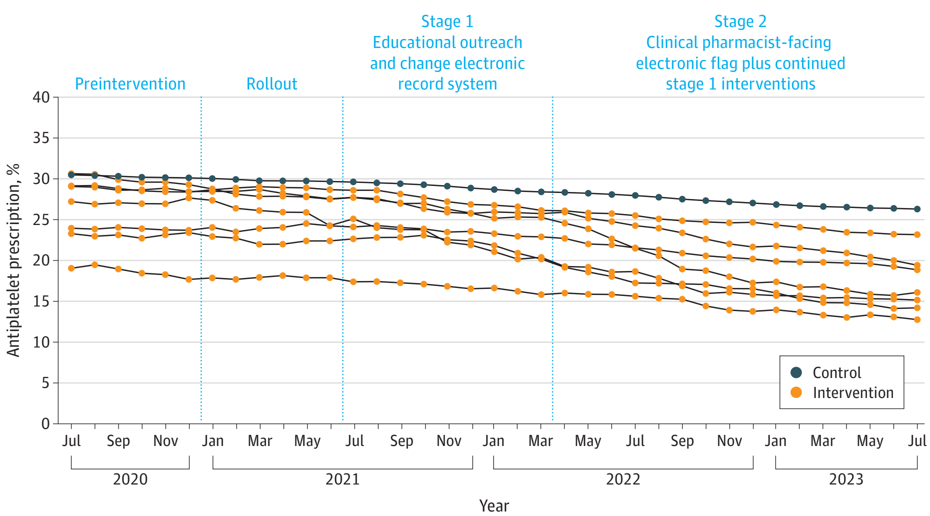
Our study adds to the literature by finding an effective multilevel approach for reducing high-risk prescribing in

Figure 2. Line Graph of the Observed Prevalence of Patients Taking Antiplatelet Medications by Clinical Subgroup and Stage of Intervention for Intervention and Control Sites



In stage 1, educational outreach and changes to the electronic health record had been implemented. In stage 2, these continued and additionally a clinical pharmacist–facing electronic flag to identify patients with DOAC-antiplatelet use was implemented. The numeric values shown in the key are from the first month of the VISN 8 intervention. Control sites consisted of all VHA sites other than the intervention sites (n = 128), with the exclusion of 10 sites that had previously used a similar antiplatelet safety intervention or had transitioned to a new electronic health record. The denominator includes all patients treated with DOAs. The "CAD event <12 mo" group includes patients with either acute coronary syndrome, percutaneous coronary intervention, or coronary artery bypass graft in the prior 12 months; such patients are recommended for antiplatelet therapy. The PAD/APL group includes patients with a prior diagnosis of peripheral artery disease or antiphospholipid syndrome. For this group, observational data are evolving and clinical trial evidence is lacking, so there are no class I guideline recommendations supporting combined DOAC-antiplatelet therapy. The stable CAD group includes patients with CAD but not classified into either of the prior 2 groups; such patients are not recommended for antiplatelet therapy if receiving therapeutic oral anticoagulation. The no-indication group includes patients not included in any of the prior groups; such patients are not recommended to receive antiplatelet therapy if receiving therapeutic oral anticoagulation. APL indicates antiphospholipid syndrome; CAD, coronary artery disease; DOAC, direct oral anticoagulant; PAD, peripheral artery disease; VHA, Veterans Health Administration; VISN, Veterans Integrated Service Network.

Figure 3. Line Graph Showing the Observed Prevalence of Antiplatelet Use by Study Stage for 7 Intervention Health Systems and Control Sites



In stage 1, educational outreach and changes to the electronic health record had been implemented. In stage 2, these continued and additionally a clinical pharmacist–facing electronic flag to identify patients with direct oral anticoagulant-antiplatelet use was implemented. Control sites (n = 128) were all Veterans Health Administration sites other than Veterans Integrated Service Network 8, with the exclusion of sites that had previously used a similar antiplatelet safety initiative or had transitioned to a new electronic health record. The denominator includes all patients treated with direct oral anticoagulants.

patients taking OACs. Few clinical interventions to reduce unnecessary combination therapy have been successful at this scale. In a prior 6-center consortium of anticoagulation services,[24] a quality improvement project involving messages to clinicians accelerated pre-existing trends in decreasing rates of antiplatelet use, with associated reductions in major bleeding. That study did not examine variation by clinical indication for antiplatelet therapy. In the primary care setting, pragmatic (tested in clinical practice) cluster-randomized quality improvement studies using multicomponent interventions, including incentive payments, clinician feedback, pharmacist support, and informatics tools, have reduced high-risk prescribing of medications such as NSAIDs and antithrombotic drugs (odds ratio, 0.58-0.63).[25,26] These studies did not specifically focus on patients taking anticoagulants. The interventions in our study had approximately similar magnitude of effect estimates as these prior studies when considering the absolute risk reduction for clinical subgroups appropriate for deprescribing. Some randomized clinical trials of discrete medication deprescribing interventions have had larger effect estimates, but these trials were smaller in scale and included more highly selected patients.[27,28]

While AD has proven effective for deimplementing other medication classes, to our knowledge this is the first study to examine AD to deimplement antiplatelet medications in this population.[11] Our study contributes to a growing evidence base for antithrombotic stewardship, which can improve OAC appropriateness, decrease bleeding, and reduce hospitalizations.[29] To put the clinical benefits in perspective, for every 12 patients with antiplatelet deimplementation, 1 major or clinically relevant nonmajor bleeding event can be prevented based on a meta-analysis of randomized trials (with a median follow-up time of 22 months).[30]

Our study is distinctive in quantifying effect estimates for 2 separate implementation approaches in the same setting. A single-center pilot study[10] intended to reduce high-risk prescribing of warfarin-antiplatelet therapy found that clinician outreach was acceptable with a high response rate, but a patient-facing education and activation tool had low levels of engagement. A meta-analysis[31] of implementation strategies targeting clinician behavior change demonstrated approximately similar effect estimates from technological implementation strategies (eg, electronic reminders) and nontechnological strategies (eg, education, local champions); however, the study did not specifically compare strategies implemented in the same setting.

It is unclear why the interventions were not more effective for patients without an indication for antiplatelet therapy, in whom the absolute effect estimates of the combined interventions were small in magnitude. The effect may have been greater in patients with stable CAD, possibly because content covered in the AD sessions was most relevant to this population.[4] Our study suggests that patients without an indication for antiplatelet therapy should be considered for future deimplementation since the group is sizable and has the most to gain from deimplementation. In addition to more clinician education on this subgroup, another option would be modifying the electronic flag to appear if and only if a patient is likely appropriate for antiplatelet deimplementation based on EHR criteria such as those used here, and further empowering clinical pharmacists to independently deimplement antiplatelet medications for this group if within their scope of practice.

There remains substantial room for improvement in the use of combination therapy. Currently, based on data from the DOAC population management tool available to pharmacists in the VHA, 22.5% of all patients taking DOACs in VHA (excluding VISN 8) also use an antiplatelet medication. In contrast, with sustained antiplatelet deimplementation, the level is now 14.9% in VISN 8, with the lowest health system at 7.6%. In VHA, National Pharmacy Benefits Management Services are examining strategies to build on and disseminate the approach evaluated here. But antiplatelet deimplementation will have to compete for clinicians' attention alongside other system priorities, such as opioid stewardship, when many clinicians already experience fatigue related to best-practice advisories and alerts.[32] In the future, continued tailoring of the interventions to the local context, including systems engineering and user-centered design approaches, could be used to minimize alert fatigue associated with electronic flags such as those in the DOAC dashboard.[33]

Replicating this intervention outside the VHA may require adaptation, especially for sites without integrated outpatient clinical pharmacists. More than 1000 academic detailers have been trained nationally,[34] and while clinical pharmacists typically provide AD in VHA, other appropriately trained individuals, including nurses, public health professionals, and advanced practice clinicians, could also fill this role. In the absence of clinical pharmacists capable of monitoring flag alerts, clinician-facing electronic clinical reminders could be sent. As an added countermeasure, payers could ensure an appropriate indication for any patient with combination therapy at the time of prior authorization for a DOAC.

Strengths and Limitations

Strengths of our study include the large sample size, the clinically meaningful comparison group, and different clinical indications studied. Some limitations are notable. First, aspirin use, although typically accurately documented in the EHR,[35] may be susceptible to misclassification because of its over-the-counter availability, which could cause overestimation or underestimation of intervention effect sizes. Second, our classification of antiplatelet indications is imperfect since we relied on documentation of care received in or paid for by VHA but did not have access to records for care covered by private insurance or Medicare.[36] However, misclassification would still be similar between intervention and control sites, which would support the validity of our findings. Third, we did not investigate clinical outcomes because of insufficient statistical power, a limitation mitigated by robust evidence of the effect estimates for combination therapy in bleeding and thrombotic outcomes. Fourth, the VHA has unique features, including predominantly male patients, dedicated anticoagulation clinical pharmacists, and more system-based capacity for quality improvement; the results may not be generalizable to all settings. Fifth, select patients with high ischemia and low bleeding risk may warrant combination therapy beyond 12 months.[37] Our methods did not allow us to identify such patients, but this scenario is uncommon and would likely have minimal effect in our principal findings. Sixth, we can only draw conclusions about the electronic flag as an augmenting strategy (ie, after the stage 1 interventions had been implemented) but not as a stand-alone intervention.

Conclusions

This quality improvement study found that a multilevel antithrombotic stewardship intervention was associated with reduced antiplatelet use in patients prescribed DOACs, particularly in patients with stable CAD, with nearly equal contributions from initial educational outreach and changes to the EHR and later augmentation by clinical pharmacist-facing electronic flags. These findings suggest that the 2 approaches have complementary roles in ensuring safe and appropriate use of high-risk antithrombotic regimens.

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Invited Commentary

LESS IS MORE

Deprescribing Inappropriate Medicines Across a Health System—Can We Improve the Care of Both Patients and Physicians?

C. Seth Landefeld, MD; Michael A. Steinman, MD

In 1860, the physician Oliver Wendell Holmes Sr told the Massachusetts Medical Society "if the whole *materia medica* … could be sunk to the bottom of the sea, it would be better for mankind and all the worse for the fishes."[1] Today, of course, the toxic medicines of 1860 have been scuttled and new medicines have been proven to cure diseases or extend life. Nonetheless, adverse drug effects remain common, and well-intentioned reminders to prevent these effects can increase physicians' cognitive burden. An especially common adverse drug effect is bleeding during anticoagulant therapy, which remains the most common drug-related cause of emergency visits.[2,3] Moreover, the number of patients prescribed an anticoagulant has increased greatly since 2000[4,5] and addition of a single antiplatelet drug, which occurs in nearly one-third of people prescribed an anticoagulant,[6] increases risk of bleeding by approximately 40%.[7,8]

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Treatment for patients with indications for both an anticoagulant and an antiplatelet has been challenging because antiplatelets even in low doses increase risk of anticoagulant-related bleeding without certainty that they reduce risk of major cardiovascular events. Recently, however, this situation has changed: compelling evidence in patients with stable coronary artery disease (CAD) shows that addition of a single antiplatelet to an anticoagulant increases major bleeding without reducing major cardiovascular events. A recent meta-